Component Stock & Job Management System

Page-wise Feature List | Laravel + MySQL

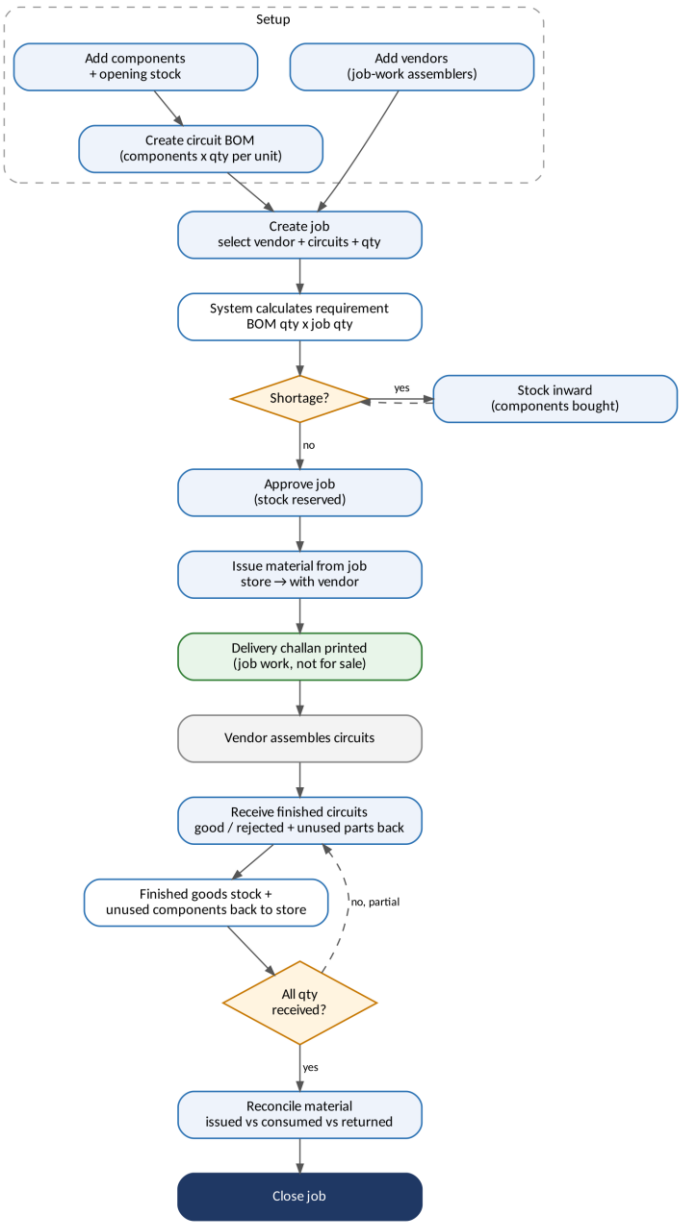
Overview

Module	Pages
1. Access	Login, Users & Roles
2. Dashboard	Dashboard
3. Masters	Component Categories, Units, Storage Locations
4. Components	Component List, Add / Edit Component, Component Details, Stock Adjustment, Bulk Import
5. Vendors (Job Work Assemblers)	Vendor List, Add / Edit Vendor, Vendor Details
6. Stock Inward	Inward List, Add Inward, Inward Details
7. Circuits (Recipes / BOM)	Circuit List, Add / Edit Circuit, Circuit Details
8. Jobs & Delivery Challan	Job List, Create / Edit Job, Job Details (with Material Issue), Delivery Challan (Print)
9. Job Receipt & Closing	Receive Finished Circuits, Job Reconciliation, Close Job, Receipt List
10. Finished Goods	Finished Goods Stock
11. Reports	Stock Report, Low Stock Report, Stock Ledger Report, Material with Vendors, Pending Receipts Report, Job Reconciliation Report, Vendor Performance Report, Job Consumption Report, Job-work Charges Report (optional), Stock Valuation, Common to all reports
12. System	Activity Log, Settings

Job Status Flow

Draft → Approved → Material Issued → Partially Received → Closed
Vendors only assemble circuits from components we provide; nothing is sold to them. A job can be Cancelled before material is issued. The vendor is selected while creating the job. Approval reserves stock, issuing material from the job generates the job-work challan and moves components to the vendor, receipt brings back finished circuits and unused parts, then the job closes.

Process Flow



1. Access

Login

- Login with email/username and password
- Remember me and forgot password (reset link by email)
- Role-based redirect after login

Users & Roles

- User list with name, role, status (active/inactive)
- Add/edit user, assign role: Admin, Store Keeper, Production
- Permission matrix per role (view/add/edit/delete/approve per module)
- Reset password and deactivate user

2. Dashboard

Dashboard

- Cards: total components, low-stock items, open jobs, finished circuits pending from vendors
- Low-stock list with quick link to add stock inward
- Jobs with material shortage
- Material lying with vendors (issued, not yet returned as finished circuits)
- Recent stock movements (last 10)
- Overdue jobs past target date, grouped by vendor

3. Masters

Component Categories

- Add/edit/delete category (e.g. Resistor, Capacitor, IC, Connector)
- Delete blocked if components are linked

Units

- Add/edit units (Nos, Meter, Reel, Pack)

Storage Locations

- Add/edit rack/bin locations for components

4. Components

Component List

- Table: part code, name, category, package, in-store stock, with-vendor qty, min stock, location
- Search and filter by category, location, low-stock only
- Low-stock rows highlighted
- Export to Excel/PDF
- Actions: view, edit, adjust stock, delete

Add / Edit Component

- Part code (unique), name, category, unit, package/footprint
- Specifications (value, tolerance, voltage, etc.)
- Minimum stock level and storage location
- Opening stock (on add only)
- Rate per unit (for valuation and circuit cost)

- Image upload and datasheet link/file

Component Details

- Summary: in-store, reserved, available, and qty lying with each vendor
- Stock ledger with date, type (inward, issue to vendor, return, adjustment), qty in/out, balance, reference
- Circuits using this component, with qty per circuit
- Inward history with rate

Stock Adjustment

- Select component, add or reduce qty
- Mandatory reason (damage, count correction, lost)
- Posted to stock ledger with user and timestamp

Bulk Import

- Download Excel template
- Upload components with opening stock
- Preview with validation errors before saving

5. Vendors (Job Work Assemblers)

Vendor List

- Vendors only assemble circuits from components we provide and return finished circuits; nothing is sold to them
- Table: name, contact person, phone, city, open jobs, pending finished qty
- Search and filter, export to Excel
- Actions: view, edit, deactivate

Add / Edit Vendor

- Name, contact person, phone, email
- GST number, address, city, state
- Circuits this vendor can assemble (optional)
- Job-work charge per unit per circuit (optional, for labour tracking)
- Remarks

Vendor Details

- Profile summary
- Open and closed jobs with ordered, received and pending qty
- Material balance: components issued, consumed, returned, and still lying with the vendor
- Delivery notes and receipts for this vendor
- Rejection and wastage summary across jobs

6. Stock Inward

Inward List

- Table: inward no, date, source/supplier name, bill no, total items, amount
- Filter by date range

Add Inward

- Enter date, supplier name (free text), bill/ref no
- Add component rows: qty, rate, amount (auto)

- Option to load items from a job shortage list
- On save, stock is added and ledger entries created
- Bill attachment upload

Inward Details

- View items and print inward entry

7. Circuits (Recipes / BOM)

Circuit List

- Table: circuit code, name, version, no. of components, material cost per unit
- Buildable qty from current stock shown per circuit
- Search and filter, export
- Actions: view, edit, duplicate, deactivate

Add / Edit Circuit

- Circuit code, name, version, description
- PCB image / layout file upload
- BOM builder: search component, add qty required per circuit, remarks (e.g. designator R1-R4)
- Duplicate component rows prevented
- Live material cost per circuit from component rates
- Import BOM from Excel

Circuit Details

- BOM table with qty per unit, available stock, and max buildable
- Bottleneck component highlighted (the one limiting buildable qty)
- Jobs created for this circuit
- Duplicate as new version

8. Jobs & Delivery Challan

Job List

- Table: job no, date, vendor, circuits, qty, status, issued/received/pending qty, target date
- Status tabs: Draft, Approved, Material Issued, Partially Received, Closed, Cancelled
- Filter by status, vendor, circuit, date range
- Quick actions: view, print challan, receive

Create / Edit Job

- Auto job number and date
- Select vendor who will assemble (required)
- Select one or more circuits with qty to prepare
- Target date for receiving finished circuits, and remarks
- Auto-calculated requirement: recipe qty x job qty, totalled across circuits
- Requirement table: required, available, shortage (highlighted)
- Job-work charge shown if set for the vendor (optional)
- Save as draft, or save and approve (reserves stock)

Job Details (with Material Issue)

- Header: job info, vendor, status, timeline of status changes
- Component table: required, reserved, issued, balance

- Issue Material button: issue qty pre-filled with balance, editable for partial issue, cannot exceed available stock
- On issue, stock moves from store to 'with vendor' and a delivery challan is generated against the job
- Multiple challans per job for partial issues, all listed on this page
- Shortage list with 'Add stock inward' action
- Receipts listed for the job, with Receive button
- Cancel job (only before issue, releases reservation)

Delivery Challan (Print)

- Challan no, date, job no, vendor name and address (from the job)
- Marked 'Material sent for job work – not for sale, to be returned as finished goods'
- Item table: component, required, issued now, total issued, balance
- Circuits and qty to be prepared from this material
- Vehicle/courier details (optional)
- Print and download PDF with signature blocks
- Challan search by number from the job list (no separate challan menu)

9. Job Receipt & Closing

Receive Finished Circuits

- Opened from job details
- Enter receipt date, vendor challan/ref no
- Per circuit: good qty and rejected qty with reason
- Vendor returns unused/excess components, added back to store
- Partial receipts allowed, status moves to Partially Received
- Finished circuits added to finished-goods stock

Job Reconciliation

- Per component: issued to vendor, consumed (received qty x recipe), returned, difference
- Missing or excess parts flagged against the vendor
- Remarks for wastage

Close Job

- Auto-close when received qty equals job qty and material is reconciled
- Manual close for short jobs with mandatory remark (Admin)
- Closed job locked from editing
- Reopen by Admin only, logged

Receipt List

- All receipts with job, vendor, date, good/rejected qty, print receipt note

10. Finished Goods

Finished Goods Stock

- Stock of finished circuits received from vendors
- Ledger: received from job, dispatched/used, adjusted

11. Reports

Stock Report

- In-store, reserved, available, with-vendor qty and value, filter by category/location

Low Stock Report

- Components below minimum level

Stock Ledger Report

- Movements by component and date range

Material with Vendors

- Component-wise qty lying with each vendor against open jobs

Pending Receipts Report

- Jobs with pending finished qty, grouped by vendor, with days pending

Job Reconciliation Report

- Wastage, rejection and shortfall per job and per vendor

Vendor Performance Report

- Jobs completed, on-time vs late, rejection %, wastage per vendor

Job Consumption Report

- Components consumed per job and per circuit

Job-work Charges Report (optional)

- Finished qty x job-work rate per vendor and date range

Stock Valuation

- Total inventory value, in store and with vendors

Common to all reports

- Date and filter options, export to Excel and PDF, print

12. System

Activity Log

- Who did what and when, filter by user, module, date

Settings

- Company name, logo, address, GST (used on challans and PDFs)
- Number prefixes for job, challan, receipt, inward
- Allow negative stock: on/off (default off)
- Low-stock alert email on/off

Use Cases

Actors: Admin, Store Keeper, Production. Each use case lists who does it, what they do, and the result in the system.

Setup

ID	Use case	Actor	Steps	Result
UC-01	**Add a new component**	Store Keeper	Enters part code, category, unit, min stock, rate and opening stock	Component available for BOMs; opening stock posted to ledger
UC-02	**Bulk import components**	Admin	Uploads Excel template, fixes validation errors, confirms	All components and opening stock created in one go
UC-03	**Add a job-work vendor**	Admin	Enters vendor details and optional job-work rate per circuit	Vendor selectable while creating jobs
UC-04	**Create a circuit BOM**	Admin / Production	Enters circuit details, adds each component with qty per unit	Recipe saved with material cost and buildable qty
UC-05	**Create a new circuit version**	Production	Duplicates existing circuit, changes a few components	New version used for new jobs; old jobs keep old BOM

Stock

ID	Use case	Actor	Steps	Result
UC-06	**Record components bought**	Store Keeper	Adds stock inward with supplier name, bill no, items, qty, rate	Store stock increases; ledger updated
UC-07	**Correct stock after physical count**	Store Keeper	Opens stock adjustment, enters qty difference and reason	Stock corrected with audit trail
UC-08	**Act on low-stock alert**	Store Keeper	Sees low-stock list on dashboard, records inward when items arrive	Items back above minimum level
UC-09	**Check how many circuits can be built**	Production	Opens circuit details	Max buildable qty and bottleneck component shown

Job and challan

ID	Use case	Actor	Steps	Result
UC-10	**Create a job for a vendor**	Production	Selects vendor, circuits and qty; reviews auto requirement	Job saved as draft with full component requirement
UC-11	**Job with shortage**	Production / Store Keeper	Job shows shortage; store records inward for short items; job re-checked	Shortage cleared, job ready to approve
UC-12	**Approve job**	Admin	Reviews requirement and approves	Stock reserved so other jobs can't use it
UC-13	**Issue material and print challan**	Store Keeper	Opens job, clicks Issue Material, confirms qty, prints challan	Stock moves to 'with vendor'; job-work challan generated
UC-14	**Partial issue**	Store Keeper	Issues only available items now, rest later from the same job	Multiple challans on one job; balance tracked

ID	Use case	Actor	Steps	Result
UC-15	**Cancel a job**	Admin	Cancels job before any material is issued	Reserved stock released

Receipt and closing

ID	Use case	Actor	Steps	Result
UC-16	**Receive finished circuits**	Store Keeper	Opens job, enters vendor challan no, good qty and rejected qty	Finished goods stock increases; pending qty reduces
UC-17	**Partial receipt**	Store Keeper	Receives part of the qty now, rest later	Job moves to Partially Received
UC-18	**Vendor returns unused components**	Store Keeper	Enters returned components during receipt	Components move from 'with vendor' back to store
UC-19	**Reconcile and close job**	Admin	Reviews issued vs consumed vs returned; system closes when all received	Job closed and locked; wastage recorded against vendor
UC-20	**Short-close a job**	Admin	Closes job with less qty received, adds mandatory remark	Job closed; shortfall shown in reconciliation report
UC-21	**Reopen a closed job**	Admin	Reopens job with reason to correct an entry	Job editable again; action logged

Monitoring

ID	Use case	Actor	Steps	Result
UC-22	**Check material lying with a vendor**	Admin	Opens vendor details or Material with Vendors report	Component-wise qty held by vendor against open jobs
UC-23	**Follow up overdue jobs**	Production	Opens pending receipts report, filters past target date	List of vendors and jobs to chase
UC-24	**Review vendor performance**	Admin	Opens vendor performance report for a period	On-time %, rejection % and wastage per vendor compared
UC-25	**Calculate job-work payment (optional)**	Admin	Opens job-work charges report for vendor and month	Finished qty x rate per vendor for payment

Key Tables

- users, roles, permissions
- categories, units, locations
- components
- vendors, vendor_circuit_rates (optional job-work charge)
- stock_inwards, stock_inward_items
- circuits, circuit_items
- jobs, job_circuits, job_items (calculated requirement)
- delivery_challans, delivery_challan_items
- job_receipts, job_receipt_items, job_receipt_returns
- stock_transactions (ledger for every stock change, with location: store or vendor)
- finished_goods, finished_goods_transactions
- activity_logs, settings